SE Lab-1

Inter-city Freight Matching Marketplace System

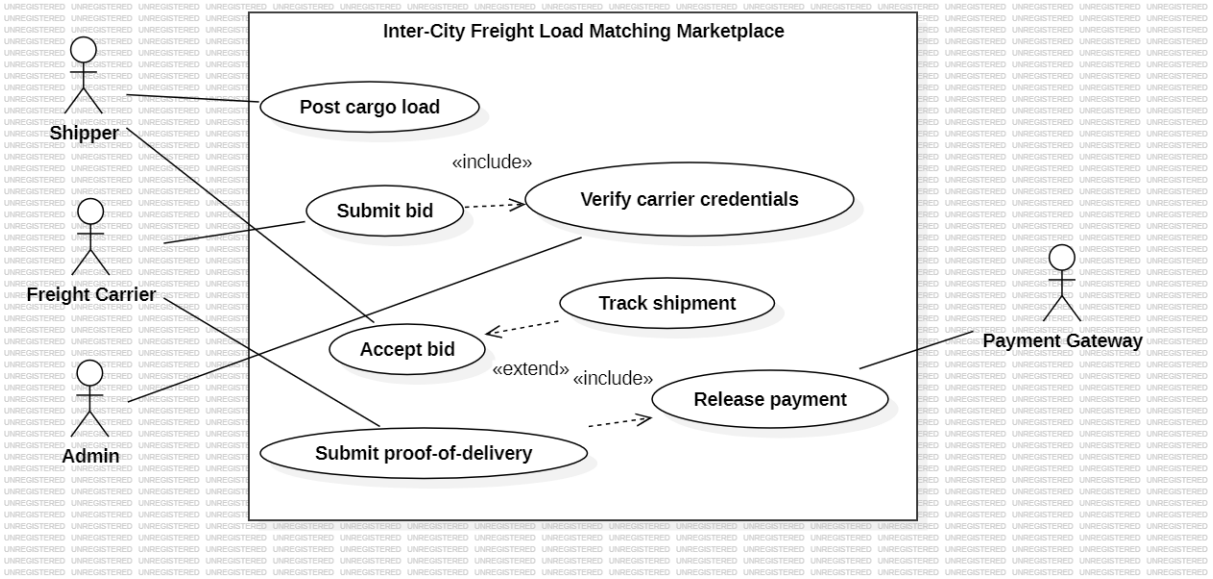
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1.Requirements Table

Req-ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow verified trucking carriers to submit competitive bids on posted freight shipments within a specified auction timeframe.	High	Carrier bid recorded and shipper notified; Fail: Unverified carrier submits bid without valid permits.	Core bidding functionality
FR-002	Functional	The system shall allow commercial shippers to post cargo loads with details within an auction timeframe.	High	Cargo load post recorded in the system and displayed to carriers.	Core marketplace posting functionality.
FR-003	Functional	The system shall allow commercial shippers to accept a bid within auction timeframe.	High	The match is confirmed and both shipper and carrier are notified.	Core bid confirmation functionality.
FR-004	Functional	The system shall allow a carrier to register a delivered load with proof-of-delivery once the delivery is complete.	High	The commercial shipper is notified and proof-of-delivery is received.	Require proof for sign-off on payment slip generation.
FR-005	Functional	The system shall automatically release milestone payment once proof-of-delivery is signed off.	High	Carrier receives milestone payment within 60 seconds and payment status for both carrier and shipper becomes ‘paid’.	Required for processing of payment to carrier.
NFR-001	Non-Functional	Milestone payments held in escrow shall disburse automatically within 60 seconds of verified electronic proof-of-delivery submission	High	Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures carrier trust
NFR-002	Non-Functional	The system shall encrypt all financial and payment verification data.	High	Security scan confirms no unencrypted financial or PII data found.	Ensures privacy of sensitive data and builds shipper and carrier trust.

2.Use Case Diagram



3.Use Case Flow Document

Goal:

Submit proof-of-delivery

Actors:

shipper, carrier, payment gateway

Preconditions:

1. A bid has been accepted for the load and a carrier has been assigned.
2. The carrier has been authenticated and verified.
3. Milestone payment funds are held in escrow.

Postconditions:

1. Funds are transferred from escrow to carriers account.
2. Shipment status is updated to ‘paid’.
3. Carrier and shipper both receive a payment complete receipt.

Main Success Scenario:

1. Carrier selects “Submit Proof-of-Delivery” in the app and uploads proof-of-delivery evidence for the shipment.
2. System time-stamps the proof-of-delivery, links it to the shipment ID, and marks the shipment “Pending Sign-Off”.
3. System sends a sign-off request notification to the Shipper.
4. Shipper reviews the POD evidence and confirms sign-off.
5. System triggers the Payment Gateway to release the escrowed milestone payment.

6. Payment Gateway processes the release to the carrier's account within 60 seconds.
7. System updates shipment status to “Paid” and generates a payment confirmation receipt.
8. System sends the confirmation receipt to both the Carrier and the Shipper
9. Use case ends successfully.

Alternate Flow:

4.Shipper disputes proof-of-delivery

4.a) Shipper flags the proof-of-delivery as disputed instead of confirming sign-off.

4.b) System marks the shipment “Disputed” and places a hold on the escrowed payment

4.c) System routes the dispute to an Admin for manual review, notifying the carrier of the hold.

4.d) Admin reviews the evidence and either releases the payment or rejects the claim and cancels the milestone payment.